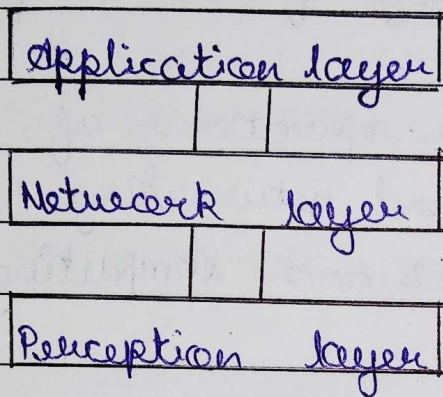
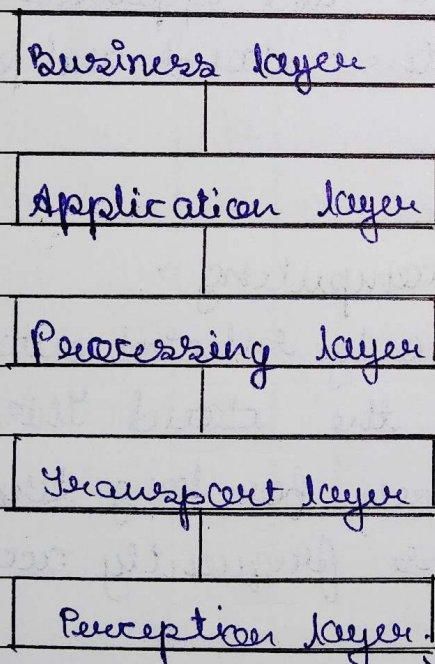


* UNIT - 2 *

- 3 Layer Structure in IoT $\Rightarrow$



- 5 Layer Structure in IoT $\Rightarrow$



$\rightarrow$ module is designed based on the requirements of the business, networks & monitoring system.

$\rightarrow$ shows the data in human language to the users, final user interface.

$\rightarrow$ decodes the data of segments by processing them, route finding.

$\rightarrow$ IP address mapping, connects packets into segments, path identification

$\rightarrow$ device selection, connectivity check, encodes the data into bits or frames.

- Cloud computing $\Rightarrow$

- **Fog computing** $\Rightarrow$ It is the term coined by Cisco that refers to extending cloud computing to an edge of the enterprise network.

$\rightarrow$ It facilitates the operations of computing storage and networking services between end devices and computing data centers.

$\rightarrow$ The fog computing all the storage capabilities, computation capabilities and data along with the applications are placed between the cloud and physical host.

when to use fog computing $\Rightarrow$

1) It is used when only selected data is required to send to the cloud. This selected data is chosen for long term storage and is less frequently accessed by the host.

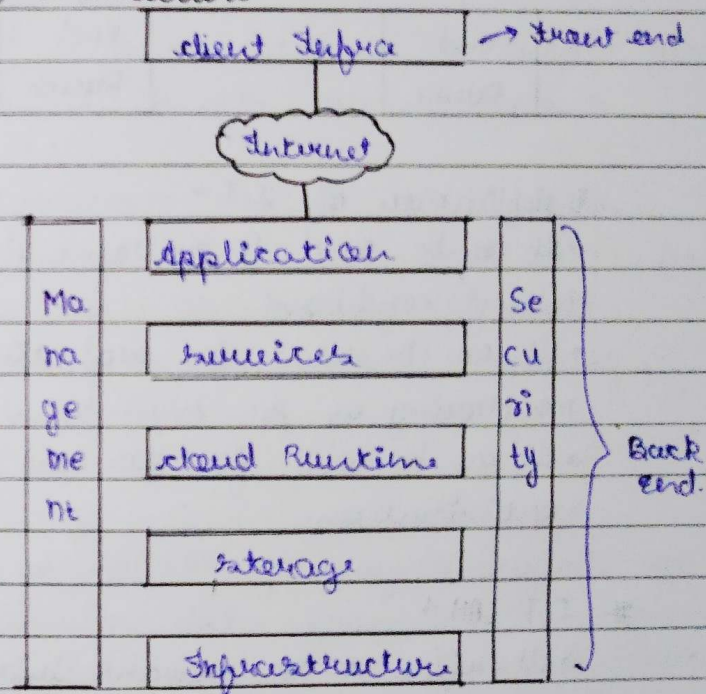
2) It is used when the data should be analyzed within a fraction of seconds i.e. latency should be low.

3) It is used whenever a large number of services need to be provided over a large area at different geographical locations.

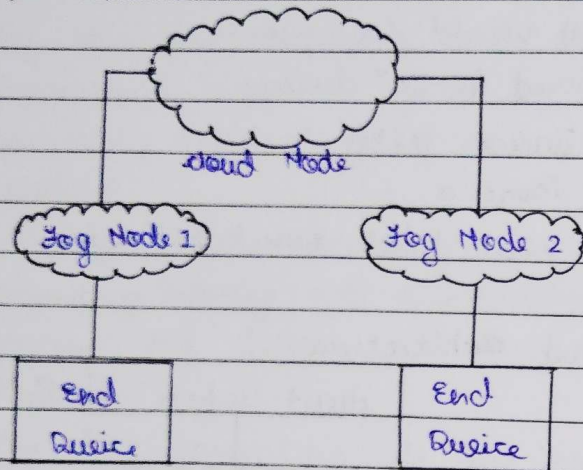
4) Real world examples:

- used in IoT devices
- Devices with sensors
- Camera
- IIOT (Industrial IoT) etc.

• cloud Architecture:



• Fog computing Archⁿ ⇒



5) Application of IoT ⇒

- It can be used to monitor and analyse the patient conditions.
- It can be used for real time rail monitoring as for high speed trains.
- It can be used for gas and oil pipeline optimisation.

IoT LAN ⇒

- It refers to the network infrastructure that connects IoT devices within a limited geographical area such as home, office or industrial environment.
- It is typically designed to facilitate communication between devices and other network subnetworks including servers, cloud services and edge

computing platforms

• components of IoT LAN ⇒

1) IoT Devices:

a) **Sensors** ⇒ It sense the quantity or input from physical environment. E.g. heat, temperature, humidity, moisture etc.
And it send respond or output gives in the human readable form.

b) **Actuators** ⇒ devices that perform action based on commands or receive data such as turning off lights, adjusting the thermostat, opening a valve (0.1)

2) Gateway:

→ In an IoT LAN gateway aggregate and transmits data from the IoT devices to higher level network system such as cloud platform or central servers.

3) Communication Protocol:

→ They are set of rules to communicate and transmit the data from source to destination. E.g.: WiFi, Bluetooth, ZigBee, Z-wave, LT, BLE (Bluetooth low energy), ethernet etc.

4) Local server & Edge devices:

→ Edge computing devices or local servers may be deployed to handle data processing analytics and decision making closer to where the data is generated.

5) Cloud Integration:

→ cloud platform are often used to store and process data collected from IoT devices in an IoT LAN.

→ The cloud provides scalable storage, advanced analytics and centralized monitoring tools.

• Benefits of IoT LAN ⇒

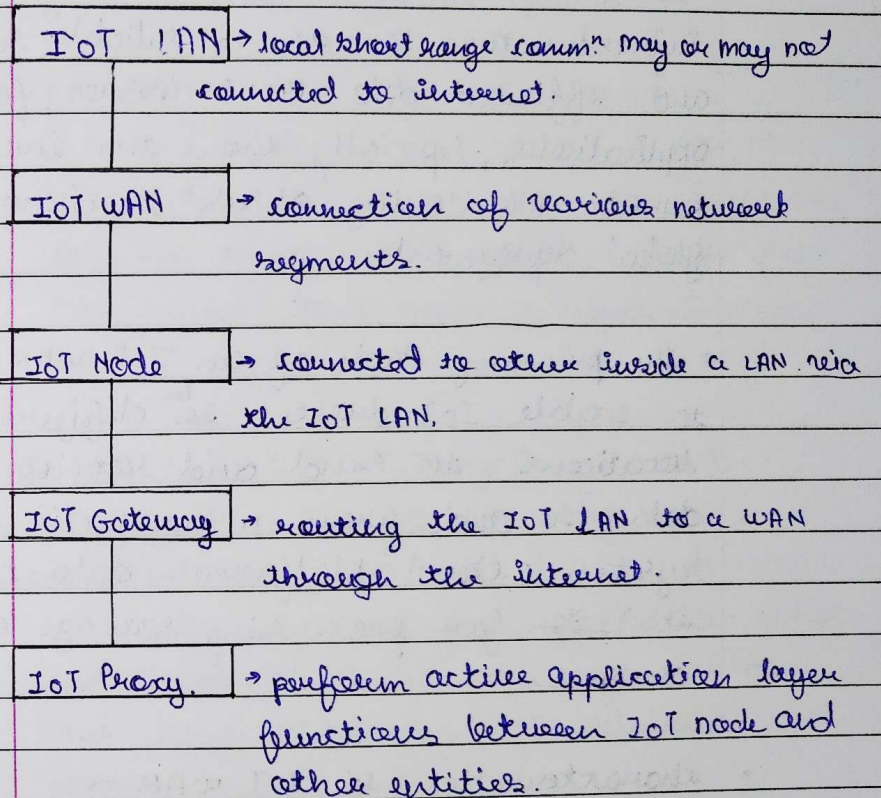
- Low latency
- Security
- Reliability
- Scalability.

• Types of IoT LAN based on network protocols ⇒

- WiFi based IoT LAN.
- Zigbi or Z-wave based IoT LAN.
- Ethernet based IoT LAN.
- BLE based IoT LAN.

IoT WAN

• Connectivity terminology of IoT WAN ⇒



→ IoT WAN refers to the type of network used to connect IoT devices over large geographical areas such as across cities, regions or even countries.

→ Unlike IoT LAN which connects devices within a limited area (like buildings, cities etc) and IoT WAN is designed to

facilitate long distance communication between IoT devices that are far apart.

→ An IoT WAN provides the necessary infrastructure to ensure reliable, scalable and efficient data transmission for IoT applications, especially those that involve remote monitoring, asset tracking or global deployments.

→ The primary role of an IoT WAN is to enable IoT devices ⁱⁿ different locations to send and receive data to and from a centralized system (cloud platforms, data centers etc.) for processing storage and analysis.

• Characteristics of IoT WAN =>

1) wide coverage area: IoT WAN are designed to cover vast geographical areas, allowing devices to communicate over distances that range from several ^{kilo-meters} kms to hundreds of kms. This is ideal for applications that need to span entire cities, rural areas or even across country.

2) Communication Protocols =>

(i) Cellular network:

→ These are widely used for IoT WAN because they offer global coverage, scalability and high data rates.

→ NB-IoT (Narrow band IoT) and LTE-M (Long term evolution for machines) are cellular standards ~~optm~~ optimized for low power long range communication in IoT applications.

(ii) ~~LoRa~~ LoRa WAN (Long Range WAN).

→ A popular low power, long range communication protocol in IoT devices.

LoRa WAN is used for applications where the devices need to communicate for ~~for~~ over long distances with minimum power consumption.

(iii) Sigfox:

→ A proprietary low power, ~~to~~ WAN (LPWAN) technology design for transmitting small amount of data over long distance. Sigfox network is used in asset tracking, remote sensing and other IoT applications.

(iv) Satellite:

→ In remote or underserved areas where terrestrial networks are unavailable satellite networks are used for IoT WAN applications offering connectivity over global distances.

(v) WiFi Mesh Network:

→ ~~wide-p~~ while primarily used for shorter range communication, WiFi mesh networks can be extended to form WAN using internet gateway to connect ^{to} disparate locations.